

Satyam Goyal

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EDUCATION

University of Michigan, Ann Arbor

Ann Arbor, MI

M.S. in Computer Science; GPA: 3.95/4.0

Expected May 2026

- Coursework: NLP, Machine Learning, Computer Vision, Databases, Parallel Programming w/ GPUs
- Conferences: AAAI 24', Y Combinator AI Startup School 25'

EXPERIENCE

IBM Research

May 2025 – Aug. 2025

Generative AI and Cloud Intern

New York, NY

- **Decreasing inference times for LLMs** hosted on distributed systems by modeling LLM serving.
- Developed an **open source vLLM simulator** that predicts industry SLO metrics with 95% accuracy.
- Used **Discrete Event Simulation** in Go and Bayesian optimizers to **decrease runtimes by 500x**.
- Published guides explaining vLLM internals to make accelerated inference serving accessible.

UMich AI Lab

Aug. 2024 – Present

Undergraduate Researcher

Ann Arbor, MI

- Developing a benchmark to evaluate agentic performance of memory over multiple conversations.
- Improved **LLM** accuracy by 18% on game-theory problems using context engineering.

Optiwise.ai Inc.

May 2024 – Jan. 2025

ML and Data Intern

Fremont, CA

- Finetuned and deployed a Qwen 30B model using **LoRA** for customized customer experience.
- Enabled data access for sales teams by implementing a Text-to-SQL AI Agent using **Langgraph**.
- Generated 2400+ qualified leads through customer data analysis using **Pandas** and **PowerBI**.
- Automated **MySQL** database management, reducing query processing time by 75%.

CodeLab - Tech Consulting Club

Oct. 2022 – Jun. 2023

Product Manager & Events Lead

Ann Arbor, MI

- Led a 9-person team to build and deploy an E-commerce product image generator with **FastAPI** and **DALL·E**, reducing seller costs by 17%.

OPEN SOURCE PROJECTS

Tailored Academic & Resource Assistant | *Python, RAG, Streamlit, VectorDB*

July 2024 – Present

- Built an **agentic chatbot** serving 150+ students for course selection and career guidance.
- Reduced response latency by 40% using optimized prompt engineering, caching, and ensemble models.
- Created documentation enabling 10+ student projects to implement similar **RAG**-based systems.

IOLBench- LLM Benchmarking | *ACL Pending*

May 2024 – Oct. 2024

- Constructed a dataset of **1,000+ linguistic problems**, enabling industry-wide model benchmarking.
- Measured performance of 8 LLMs across language and time-series tasks using an evaluation pipeline.
- Created a **Python library** with **NumPy** and **RegEx** for scaling model evaluations of LLMs.

TECHNICAL SKILLS

Languages (Proficient): Python, SQL

Languages (Familiar): Java, C++, JavaScript, Go

ML/AI/Data: PyTorch, RAG, Scikit-Learn, Agentic AI, Transformers, PostgreSQL, Pandas

Cloud & DevOps: vLLM, LM Cache, AWS (EC2, Lambda, SES), Docker, Kubernetes, FastAPI